



Software Requirements Specification

for

Video Rental Store Management System (VRSMS)

Version 1.2

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Revision History

Version	Date	Description of Changes
1.0	February 06, 2026	Initial submission (Desktop Application focus).
1.1	February 10, 2026	Major Update: Re-architected as Web-Based System with Member Portal and OTP Auth.
1.2	February 16, 2026	Refined Business Logic: Removed Online Payments and Cart. Replaced with Wishlist. Added 10-day loan limit, overdue fines, and auto-cancellation rules. Added Manager Configuration for numerical parameters.

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1 Introduction

1.1 Purpose

The purpose of this Software Requirements Specification (SRS) is to describe the functional and non-functional requirements for Release 1.2 of the Video Rental Store Management System (VRSMS). This document is intended to be used by the members of the project team (designers, developers, testers) to implement and verify the correct functioning of the system. It also serves as a validation agreement between the development team and the Store Manager.

1.2 Project Scope and Product Features

1.2.1 Project Overview

The Video Rental Store is in need of an automated management system to replace its current manual record-keeping. The store deals in a large volume of media, including Video CDs, DVDs (VHS/MP4 formats), and Music CDs. Currently, operations such as tracking membership deposits, calculating rental fees based on item age, and identifying “dead stock” for sale are done manually.

The product will be a **Web-Based Application** hosted on a centralized server. This architecture ensures that data is accessible not only to store staff via counter terminals but also to registered members via personal devices. The system will facilitate real-time inventory checking, wishlisting, and strict enforcement of rental duration limits.

1.2.2 Project Scope

The VRSMS logic is strictly defined by the store’s business rules. The system scope allows for:

1. **Membership Administration:** Registering members with Photo ID and OTP verification.
2. **Configurable Business Logic:** Allowing the Manager to set parameters for rental duration (default 10 days), overdue fines, and cancellation thresholds (default 20 days).
3. **Inventory Tracking:** Managing the catalog with automated rate adjustments and identifying “dead stock.”
4. **Online Catalog Wishlist:** Members can browse inventory and save items to a personal Wishlist (without affecting stock availability).
5. **Circulation (Borrow/Return):** Enforcing limits, calculating dues (Rent + Fines), and handling auto-cancellation of defaulters.
6. **Remote Reporting:** Providing the Manager with remote access to Profit/Loss reports.

1.3 References

1. Problem Statement: “Video Rental Store Management System (VRSMS)”, Shiv Nadar University, Course Material.
2. IEEE Std 830-1998: “IEEE Recommended Practice for Software Requirements Specifications”.
3. Sample SRS: “Software Requirements Specification for RIT Peer Evaluation System”, Team Green Apple.

1.4 Definitions, Acronyms, and Abbreviations

Term	Definition
SRS	Software Requirements Specification
VRSMS	Video Rental Store Management System
OTP	One-Time Password (used for authentication)
Dead Stock	Inventory items that have not been issued for more than 365 days
Wishlist	A personal list of items a user intends to rent. Does not reserve stock.

1.5 Organization of the SRS

The remainder of this document is organized as follows: Section 2 provides an overview of the product, user characteristics, and operating environment. Section 3 details the specific functional requirements arranged by system feature. Section 4 defines the external interface requirements including user, hardware, and software interfaces. Section 5 specifies non-functional requirements.

2 Overall Description

2.1 Product Perspective

The VRSMS is a web-based software solution. It consists of a centralized database and a web application server.

- **Store Terminals:** Accessed by Clerks/Managers via web browser on store computers for transactions.
- **Remote Clients:** Accessed by Members via browsers on personal devices for browsing/profile management.
- **Database:** A centralized SQL-based cloud database.

2.2 User Classes and Characteristics

User Class	Description
Store Clerk	The primary front-desk user. Responsibilities include registering members, processing physical handovers (borrow/return), and collecting payments at the store counter.
Store Manager	The administrative user. Responsibilities include inventory control, configuring system parameters (fines/durations), and viewing financial reports.
Member	A registered customer. Can log in remotely using OTP to browse the catalog, manage their Wishlist, and check outstanding dues.

2.3 Operating Environment

- **OE-1:** Server: Linux-based Cloud Server (AWS/Azure) or Local Intranet Server.
- **OE-2:** Client: Any modern Web Browser (Chrome, Firefox, Edge).
- **OE-3:** Database: MySQL or PostgreSQL.

2.4 Design and Implementation Constraints

- **CO-1:** Internet connectivity is required for all operations.
- **CO-2:** User authentication must be performed via OTP (No static passwords for Members).
- **CO-3:** Payments are strictly in-store; the web portal is for information only.

2.5 Assumptions and Dependencies

- **AS-1:** It is assumed the store has a stable internet connection.
- **AS-2:** It is assumed the Store Manager will configure the fine/duration parameters upon system initialization.

3 System Features

3.1 Manage Membership (Registration & Auth)

3.1.1 Description and Priority

High Priority. The system allows the Clerk to register new members. Security is enhanced by capturing the user's photo and verifying their phone number via OTP.

3.1.2 Stimulus/Response Sequences

- **Stimulus:** Clerk initiates "New Member Registration".
- **Response:** System requests Name, Address, and Phone Number.
- **Stimulus:** Clerk enters phone number and requests verification.
- **Response:** System sends a 6-digit OTP to the user's phone.
- **Stimulus:** Clerk enters the OTP provided by the user.
- **Response:** System verifies OTP. System prompts Clerk to capture User Photo via web-cam.
- **Stimulus:** Clerk submits details and Deposit Payment reference (Cash/Card/UPI collected at store).
- **Response:** System creates the account.

3.1.3 Functional Requirements

Feature ID	Feature Name	Description
FR-MEM-01	OTP Verification	The system shall generate and verify a One-Time Password (OTP) for phone number validation during registration and login.
FR-MEM-02	Photo Capture	The system shall interface with a camera to capture and store a photo of the member during registration.
FR-MEM-03	Register Member	The application shall record member details and the Rs. 1000 deposit.

3.2 Inventory Management & Online Catalog

3.2.1 Description and Priority

High Priority. Allows the Manager to manage stock and Members to browse availability remotely.

3.2.2 Stimulus/Response Sequences

- **Stimulus:** Member searches for “The Matrix” from their home computer.
- **Response:** System searches the database. Status is shown as “Available” based strictly on physical stock, or “On Loan” with expected return date.
- **Stimulus:** Member clicks “Add to Wishlist”.
- **Response:** System saves the item to the user’s personal list. Availability status of the item remains unchanged.

3.2.3 Functional Requirements

Feature ID	Feature Name	Description
FR-INV-01	Online Search	Members shall be able to filter inventory by Category (Video/Music), Genre, and Format (VHS/MP4).
FR-INV-02	Real-Time Status	The system shall display status as “Available” or “On Loan”.
FR-INV-03	Wishlist	The system shall allow members to save items to a Wishlist. This action shall not reserve the item or affect inventory counts.
FR-INV-04	Auto-Halving Rate	The system shall halve the daily rental rate for items purchased > 365 days ago.
FR-INV-05	Dead Stock Report	The system shall identify items unissued for > 1 year for the Manager.

3.3 Process Rentals (Borrow/Return)

3.3.1 Description and Priority

High Priority. Handles physical exchange of items, enforcement of loan durations, and calculation of overdue fines.

3.3.2 Stimulus/Response Sequences

- **Stimulus:** Clerk scans an item for a Member.
- **Response:** System checks availability and member limits (1 Video + 1 Music). System records issue date.
- **Stimulus:** Item is held for more than the configured Max Days (Default: 10).
- **Response:** System begins accumulating an extra “Overdue Fine” per day.
- **Stimulus:** Item is held for more than the Cancellation Threshold (Default: 20 days).
- **Response:** System automatically marks the membership as “Cancelled” and alerts the Manager.
- **Stimulus:** Clerk processes return.
- **Response:** System calculates Total Dues = (Rent × Days) + (Overdue Fine × Overdue Days).

3.3.3 Functional Requirements

Feature ID	Feature Name	Description
FR-REN-01	Enforce Limits	The system shall restrict loans to Max 1 Video + 1 Music per member simultaneously.
FR-REN-02	Loan Duration	The system shall track the number of days an item is held.
FR-REN-03	Overdue Fines	The system shall apply a configurable extra fine per day if the loan duration exceeds the configured limit (Default: 10 days).
FR-REN-04	Auto-Cancellation	The system shall automatically cancel a membership if an item is not returned within the threshold (Default: 20 days).

3.4 Managerial Configuration & Remote Access

3.4.1 Description and Priority

Medium Priority. Allows the Manager to configure business rules and view reports remotely.

3.4.2 Functional Requirements

Feature ID	Feature Name	Description
FR-MGR-01	Remote Dashboard	The system shall allow the Manager to login remotely and view “Profit/Loss” and “Dead Stock” reports.
FR-MGR-02	System Configuration	The system shall allow the Manager to modify numerical parameters: Max Loan Duration, Overdue Fine Amount, and Cancellation Threshold.

3.5 Member Self-Service Portal

3.5.1 Description and Priority

Medium Priority. A dashboard for members to view their data and Wishlist.

3.5.2 Stimulus/Response Sequences

- **Stimulus:** Member requests login on the web portal.
- **Response:** System requests Phone Number → Sends OTP → Verifies OTP → Grants Access.
- **Stimulus:** Member views “Fine Tracker”.
- **Response:** System displays total outstanding dues (Rent + Fines). Note: Payment must be made at the store.

3.5.3 Functional Requirements

Feature ID	Feature Name	Description
FR-WEB-01	Remote Login	The system shall authenticate members via OTP.
FR-WEB-02	Loan History	The system shall display a list of all current and past borrowed items.
FR-WEB-03	Fine Tracker	The system shall display outstanding dues.
FR-WEB-04	Manage Wishlist	The system shall allow users to add or remove items from their personal Wishlist.

4 External Interface Requirements

4.1 User Interfaces

- **UI-1: Web Portal:** Responsive web interface accessible on Desktop and Mobile browsers.
- **UI-2: Dashboards:** Distinct views for Clerk (POS/Transaction focus), Manager (Admin/Config focus), and Member (Catalog/Profile focus).

4.2 Hardware Interfaces

- **HW-1: Webcam:** Connected to the Clerk's terminal for capturing Member photos during registration.
- **HW-2: Receipt Printer:** Thermal printer for physical transaction slips.

4.3 Software Interfaces

- **SI-1: SMS Gateway:** Integration with a third-party SMS provider for sending OTPs.
- **SI-2: Database:** Cloud-hosted SQL Database.

4.4 Communications Interfaces

- **CI-1: Internet:** The system requires a continuous HTTPS connection to the central server.

5 Other Nonfunctional Requirements

5.1 Performance Requirements

- **PE-1:** Online catalog search results must load within 2 seconds.
- **PE-2:** OTP delivery should occur within 10 seconds of request (dependent on SMS gateway).

5.2 Security Requirements

- **SE-1:** All data transmission must be encrypted via HTTPS/SSL.
- **SE-2:** Member passwords are replaced by OTP authentication to prevent unauthorized access.
- **SE-3:** Managerial configuration settings must require a secondary admin login.